In-medium similarity renormalization group at finite temperature

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Abstract

The study of nuclei at finite temperature is of immense interest for many areas of nuclear astrophysics and nuclear-reaction science. A variety of *ab initio* methods are now available for computing the properties of nuclei from interactions rooted in quantum chromodynamics, but applications have largely been limited to zero temperature. In the present work, we extend one such method, the in-medium similarity renormalization group (IMSRG), to finite temperature. Using an exactly solvable schematic model that captures essential features of nuclear interactions, we show that the FT-IMSRG can accurately determine the energetics of nuclei at finite temperature, and we explore the accuracy of the FT-IMSRG in different parameter regimes, e.g., strong and weak pairing. In anticipation of FT-IMSRG applications for finite nuclei and infinite matter, we discuss differences arising from the choice of working with the canonical and the grand-canonical ensembles. In future work, we will apply the FT-IMSRG with realistic nuclear interactions to compute nuclear structure and reaction properties at finite temperature, which are important ingredients for understanding nucleosynthesis in stellar environments, or modeling reactions of hot compound nuclei.

本文章将IMSRG方法拓展至有限温度情况下，并基于可严格求解的P3H模型评估其准确性。

Introduction

- 从头计算方法是基于第一性原理研究量子多体问题的重要方法，在研究原子核性质取得了不错的成功，其中包括Hartree-Fock方法以及IMSRG方法。
- 迄今为止，IMSRG及其他超越平均场方法在有限核体系中的应用仍局限于零温条件，尽管基于现代核相互作用的有限温度无限核物质研究已通过多种方法实现，例如Finite Temperature(FT)-HF, FT-HFB方法等，但这些方法依赖的是参数拟合的哈密顿量或有效相互作用。
- 在这项工作中，我们提出了有限温度IMSRG (FT-IMSRG) 方法，并通过精确可解的P3H模型评估其精确度。我们的目标是为有限温度下核性质、衰变和反应速率的从头计算奠定基础。这些研究对于理解高温恒星环境中的核合成过程或热核反应至关重要

In-Medium Similarity Renormalization Group

Flow equation：

$$H(s) \equiv U(s) H U^{\dagger}(s),$$

$$\frac{d}{ds} U(s) = \eta(s) U(s).$$

$$\frac{d}{ds} H(s) = [\eta(s), H(s)].$$

Normal-ordered Hamiltonian

$$\hat{H} = E + \sum_{ij} f_{ij} \{a_i^{\dagger} a_j\} + \frac{1}{4} \sum_{ijkl} \Gamma_{ijkl} \{a_i^{\dagger} a_j^{\dagger} a_l a_k\}$$

IMSRG(2) flow equation:

$$\frac{d}{ds} E = \sum_{ab} \eta_{ab} f_{ab} (n_a - n_b) + \frac{1}{2} \sum_{abcd} \eta_{abcd} f_{cdab} n_a n_b \bar{n}_c \bar{n}_d$$

$$\begin{aligned} \frac{df_{ij}}{ds} = & \sum_{a} (1 + P_{ij}) \eta_{ia} f_{aj} + \sum_{ab} (n_a - n_b)(\eta_{ab} \Gamma_{biaj} - f_{ab} \eta_{biaj}) \\ & + \frac{1}{2} \sum_{abc} (n_a n_b \bar{n}_c + \bar{n}_a \bar{n}_b n_c)(1 + P_{ij}) \eta_{ciab} \Gamma_{abcj} \end{aligned}$$

$$\begin{aligned} \frac{d\Gamma_{ijkl}}{ds} = & \sum_{a} \{(1 - P_{ij})(\eta_{ia} \Gamma_{ajkl} - f_{ia} \eta_{ajkl}) - (1 - P_{kl})(\eta_{ak} \Gamma_{ijal} - f_{ak} \eta_{ijal})\} \\ & + \frac{1}{2} \sum_{ab} (1 - n_a - n_b)(\eta_{ijab} \Gamma_{abkl} - \Gamma_{ijab} \eta_{abkl}) \\ & + \sum_{ab} (n_a - n_b)(1 - P_{ij})(1 - P_{kl}) \eta_{aibk} \Gamma_{bjal} \end{aligned}$$

Finite-temperature Hartree-Fock

$\beta \equiv 1/T \quad (\text{with } k_B = 1)$

	grand-canonical ensemble	canonical ensemble
Density operator	$\hat{\rho} \equiv \frac{1}{Z} e^{-\beta(\hat{H}-\mu\hat{A})}$	$\hat{\rho}_A \equiv \frac{1}{Z_A} e^{-\beta\hat{H}}$
Partition function	$Z \equiv \mathrm{Tr} e^{-\beta(\hat{H}-\mu\hat{A})}.$	$Z_A \equiv \mathrm{Tr} e^{-\beta\hat{H}}$
Occupation numbers	$n_p = \mathrm{Tr}(\hat{\rho} a_p^\dagger a_p) = \frac{1}{1+e^{\beta(e_p-\mu)}}, \quad \bar{n}_p \equiv 1 - n_p.$ $\sum_p n_p = A$	$n_p = \frac{Z_{A-1}^{\backslash\{p\}} e^{-\beta\epsilon_p}}{Z_A},$
Feature	计算简单； 允许系统和外界交换能量和粒子，粒子平均数为A，存在粒子数涨落	计算复杂； 允许系统和外界交换能量，粒子数守恒，更适合原子核系统

➢ 由于使用的模型较为简单，本文章使用正则系综计算粒子占据数

FT-HF iteration procedure

We begin by expressing the Hamiltonian as
$$\hat{H} = \hat{K} + \hat{V} = \sum_{\alpha\beta} k_{\alpha\beta} a_\alpha^\dagger a_\beta + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} v_{\alpha\beta\gamma\delta} a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma ,$$

$$\hat{f}^{(0)} = \hat{k}.$$

(1) Diagonalize $\hat{f}$, yielding eigenvalues $\{\epsilon_p\}$ and a unitary transformation $\hat{u}$.

(2) Calculate the (grand-)canonical occupation numbers n_p based on the eigenvalues $\{\epsilon_p\}$.

(3) Construct the density matrix via Eq. (42). $\rho_{\alpha\beta}^{(i)} = \sum_p n_p^{(i)} u_{\alpha p}^{(i)} u_{\beta p}^{(i)},$

(4) Construct a new $\hat{f}$ and HF energy E using Eqs. (43) and (44).

$$f_{\alpha\beta}^{(i+1)} = k_{\alpha\beta} + \sum_{\gamma\delta} v_{\alpha\gamma\beta\delta} \rho_{\gamma\delta}^{(i)}, \quad E^{(i+1)} = \sum_{\alpha\beta} k_{\alpha\beta} \rho_{\alpha\beta}^{(i)} + \frac{1}{2} \sum_{\alpha\beta\gamma\delta} v_{\alpha\beta\gamma\delta} \rho_{\alpha\gamma}^{(i)} \rho_{\beta\delta}^{(i)}.$$

$$\frac{|E^{(i+1)} - E^{(i)}|}{A} \leqslant 10^{-5}.$$

NO

YES

$$K_{pq} = \sum_{\alpha\beta} U_{p\alpha}^* k_{\alpha\beta} U_{\beta q}$$

$$V_{pqrs} = \sum_{\alpha\beta\gamma\delta} U_{p\alpha}^* U_{q\beta}^* v_{\alpha\beta\gamma\delta} U_{\gamma r} U_{\delta s}.$$

For FT-IMSRG

Finite-temperature IMSRG

- Using **the grand-canonical FT-HF** partition function and density operator, the normal ordering and Wick's theorem generalize to finite temperature in a straightforward way.

$$\overline{a_p^\dagger a_q} \equiv \langle a_p^\dagger a_q \rangle = n_p \delta_{pq} \,,$$

$$\overline{a_p a_q^\dagger} \equiv \langle a_p a_q^\dagger \rangle = \bar{n} \delta_{pq} \,,$$

$$\begin{aligned} \langle a_p^\dagger a_q^\dagger a_s a_r \rangle &= \langle a_p^\dagger a_r \rangle \langle a_q^\dagger a_s \rangle - \langle a_p^\dagger a_s \rangle \langle a_q^\dagger a_r \rangle \\ &= n_p n_q (\delta_{pr}\delta_{qs} - \delta_{ps}\delta_{qr}), \end{aligned}$$

- Even though the **canonical FT-HF** Hamiltonian and partition function still only involve up to one-body operators, Wick's theorem must be amended with additional terms that ensure that the particle number is fixed.

- **generator**

$$\begin{aligned} \hat{\eta} = \; & \frac{1}{2} \sum_{pq} \bar{n}_p n_q \arctan\left(\frac{2 f_{pq}}{\Delta_{pq}}\right) \{a_p^\dagger a_q\} \\ & + \frac{1}{8} \sum_{pqrs} \bar{n}_p \bar{n}_q n_r n_s \arctan\left(\frac{2\Gamma_{pqrs}}{\Delta_{pqrs}}\right) \{a_p^\dagger a_q^\dagger a_s a_r\} - \text{H.c.}, \end{aligned}$$

$$\begin{aligned} \Delta_{pq} &= \langle \{a_q^\dagger a_p\} \hat{H} \{a_p^\dagger a_q\} \rangle - \langle \hat{H} \rangle \\ &= \bar{n}_p n_q (\bar{n}_p f_{pp} - n_q f_{qq} - \bar{n}_p n_q \Gamma_{pqpq}) \end{aligned}$$

$$\begin{aligned} \Delta_{pqrs} &= \langle \{a_r^\dagger a_s^\dagger a_q a_p\} \hat{H} \{a_p^\dagger a_q^\dagger a_s a_r\} \rangle - \langle \hat{H} \rangle \\ &= \bar{n}_p \bar{n}_q n_r n_s (\bar{n}_p f_{pp} + \bar{n}_q f_{qq} - n_r f_{rr} - n_s f_{ss} \\ &\quad + \bar{n}_p \bar{n}_q \Gamma_{pqpq} + n_r n_s \Gamma_{rsrs} - \bar{n}_p n_r \Gamma_{prpr} \\ &\quad - \bar{n}_q n_s \Gamma_{qsqs} - \bar{n}_p n_s \Gamma_{psps} - \bar{n}_q n_r \Gamma_{qrqr}). \end{aligned}$$

Particle number variance

- The particle number operator

$$\hat{A} = \sum_p a_p^\dagger a_p$$

- It can drive that

$$\langle \hat{A} \rangle = A$$

$$\langle \hat{A}^2 \rangle = A^2 + \sum_p \bar{n}_p n_p,$$

$$\Delta A = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 = \sum_p \bar{n}_p n_p.$$

- While it appears that this result is in full generality, the canonical ensemble is defined such that ΔA = 0. The previously mentioned corrections to Wick's theorem in the canonical ensemble enforce the condition ΔA = 0.

$$[\hat{\eta}, \hat{A}]^{0B} = \sum_{pq} (n_p - n_q)\eta_{pq}\delta_{pq} = 0,$$

$$[\hat{\eta}, \hat{A}]^{1B}_{pq} = \sum_r (1 + P_{pq})\eta_{pr}\delta_{rq} - \sum_{rs} (n_r - n_s)\delta_{rs}\eta_{sprq}$$
$$= \eta_{pq} + \eta_{qp} = 0$$

$$[\hat{\eta}, \hat{A}]^{2B}_{pqrs} = -\sum_t ((1 - P_{pq})\delta_{pt}\eta_{tqrs} - (1 - P_{rs})\delta_{tr}\eta_{pqts})$$
$$= -(\eta_{pqrs} - \eta_{qprs} + \eta_{pqrs} - \eta_{pqsr}) = 0. \quad (7$$

$$\frac{d}{ds}\hat{A} = [\hat{\eta}(s), \hat{A}] = 0. \quad [\hat{\eta}, \hat{A}^2] = \hat{A}[\hat{\eta}, \hat{A}] + [\hat{\eta}, \hat{A}]\hat{A} = 0.$$

- The average particle number as well as the particle number variance are invariant under the FT-IMSRG flow andentirely determined by the reference ensemble

P3H model

The Hamiltonian of P3H model

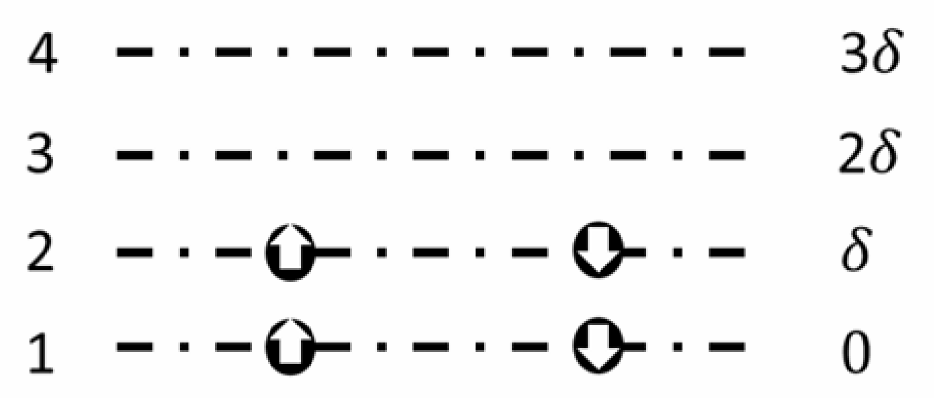
$$\hat{H} = \hat{K} + \hat{V} = \sum_{\alpha\beta} k_{\alpha\beta} a_\alpha^\dagger a_\beta + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} v_{\alpha\beta\gamma\delta} a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma ,$$

$$\hat{K}_{\mathrm{P3H}} = \delta \sum_{\alpha\sigma} (\alpha - 1) a_{\alpha\sigma}^\dagger a_{\alpha\sigma}$$

$$\hat{V}_{\mathrm{P3H}} = -\frac{g}{2} \sum_{\alpha\beta} a_{\alpha+}^\dagger a_{\alpha-}^\dagger a_{\beta-} a_{\beta+}$$

$$- \frac{b}{2} \sum_{\alpha,\beta,\gamma\neq\beta} (a_{\alpha+}^\dagger a_{\alpha-}^\dagger a_{\beta-} a_{\gamma+} + a_{\gamma+}^\dagger a_{\beta-}^\dagger a_{\alpha-} a_{\alpha+}).$$

Schematic diagram



- Here δ is the (constant) spacing between single-particle energy levels, g is the strength of pairing interaction, and b controls the strength of pair-breaking, particle-hole type excitations.

Result

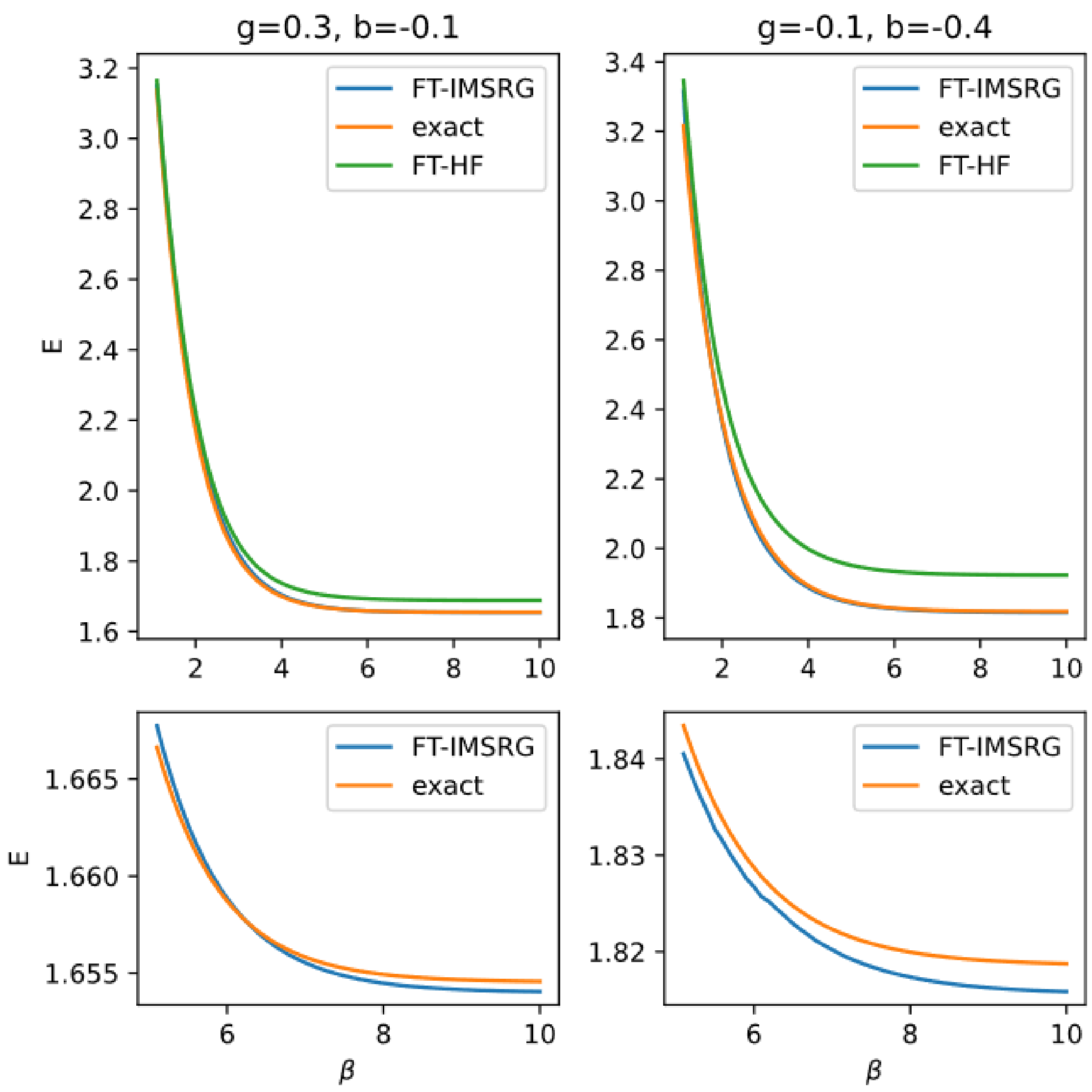


FIG. 2. Internal energy versus inverse temperature with $A = 4$ and $N = 8$ for $g = 0.3$, $b = -0.1$ (left) and $g = -0.1$, $b = -0.4$ (right). The bottom panels show the low-temperature range, $\beta \geqslant 5.0$. The FT-IMSRG results (blue) are much closer to the exact results (orange) than the FT-HF results (green) are. The FT-IMSRG results successfully replicate the behavior of the exact results at low temperatures.

- 相比于FT-HF，FT-IMSRG更接近严格解，尤其是在低温情况下(较大β)。

$$\beta \equiv 1/T$$

Result

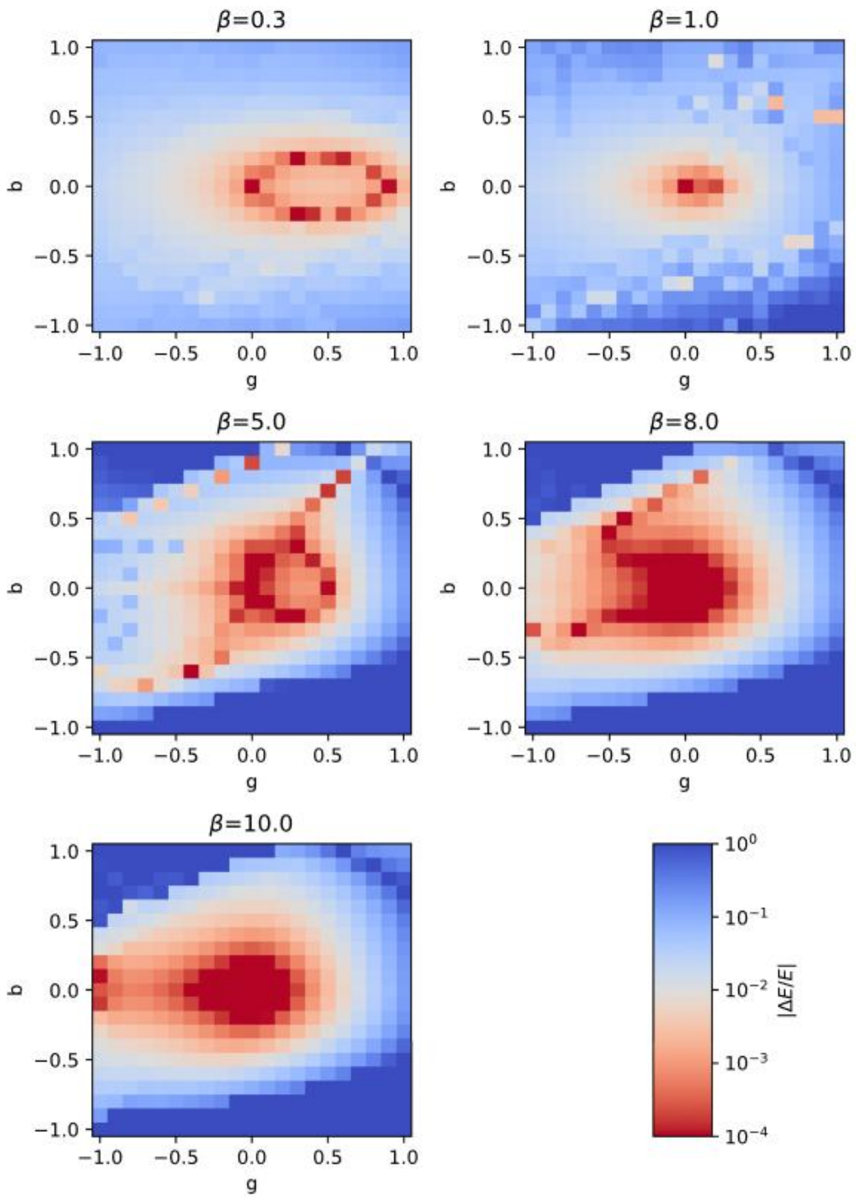


FIG. 3. Relative error in internal energy in parameter space for various inverse temperatures with a logarithmic color scale. Shades of red denote less than 1% error. The FT-IMSRG results display the most accuracy for lower temperatures and parameters of lower magnitude.

- 当|g|和|b| ≤ 0.5时，FT-IMSRG的计算结果与精确解呈现良好的一致性。
- FT-IMSRG方法在中等温度范围（约β=1）表现最弱，随着β趋近于零，其性能有所改善。这是因为在这些温度下，更广泛的单粒子态具有显著的非零占据数，因此解耦条件的要求变得更加严格。
- 当|g|和|b|变得过大时，尤其是当它们符号相反时，FT-IMSRG方法经常会出现发散现象。这是因为正的g会促进配对，而负的b会抑制配对破坏，反之亦然，从而导致相互增强效应。

Result

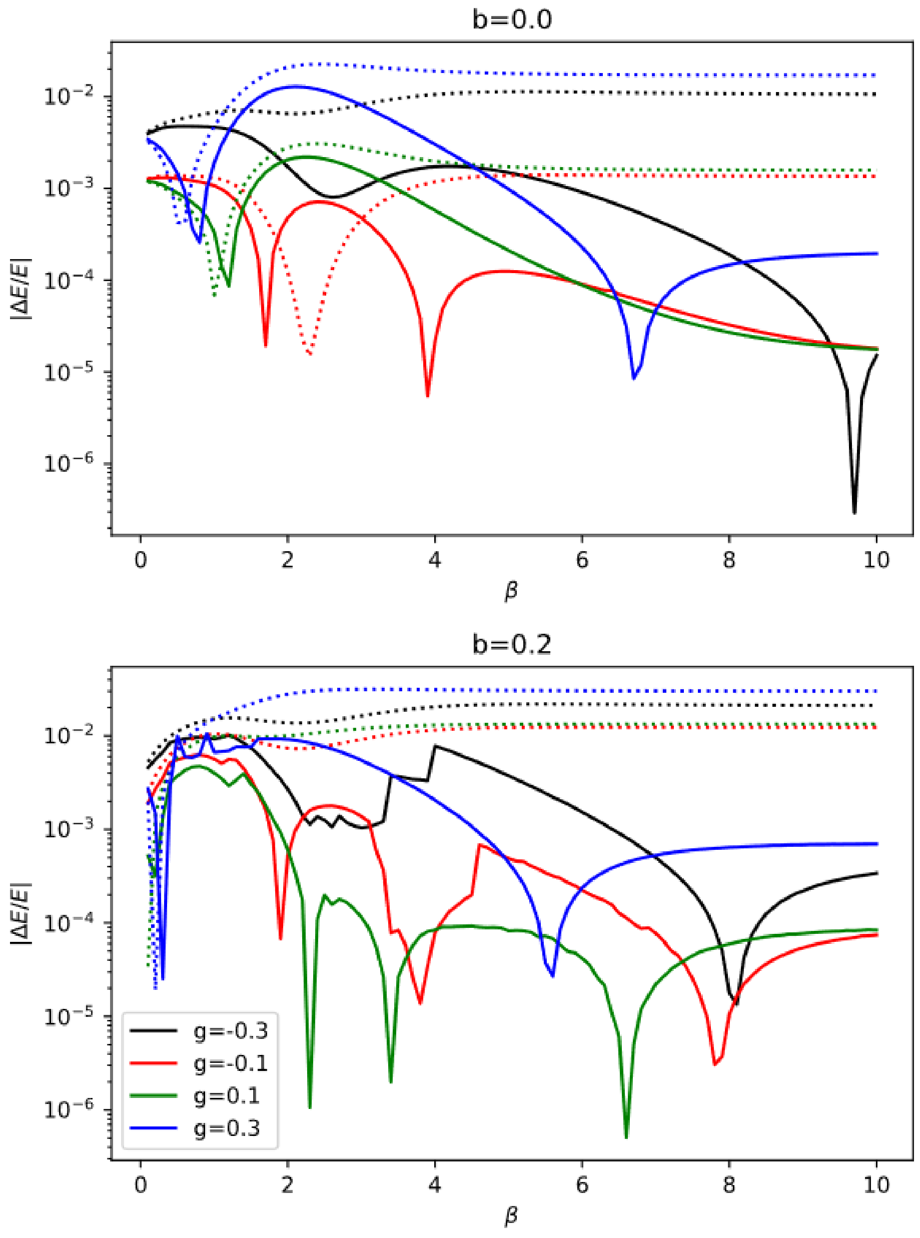
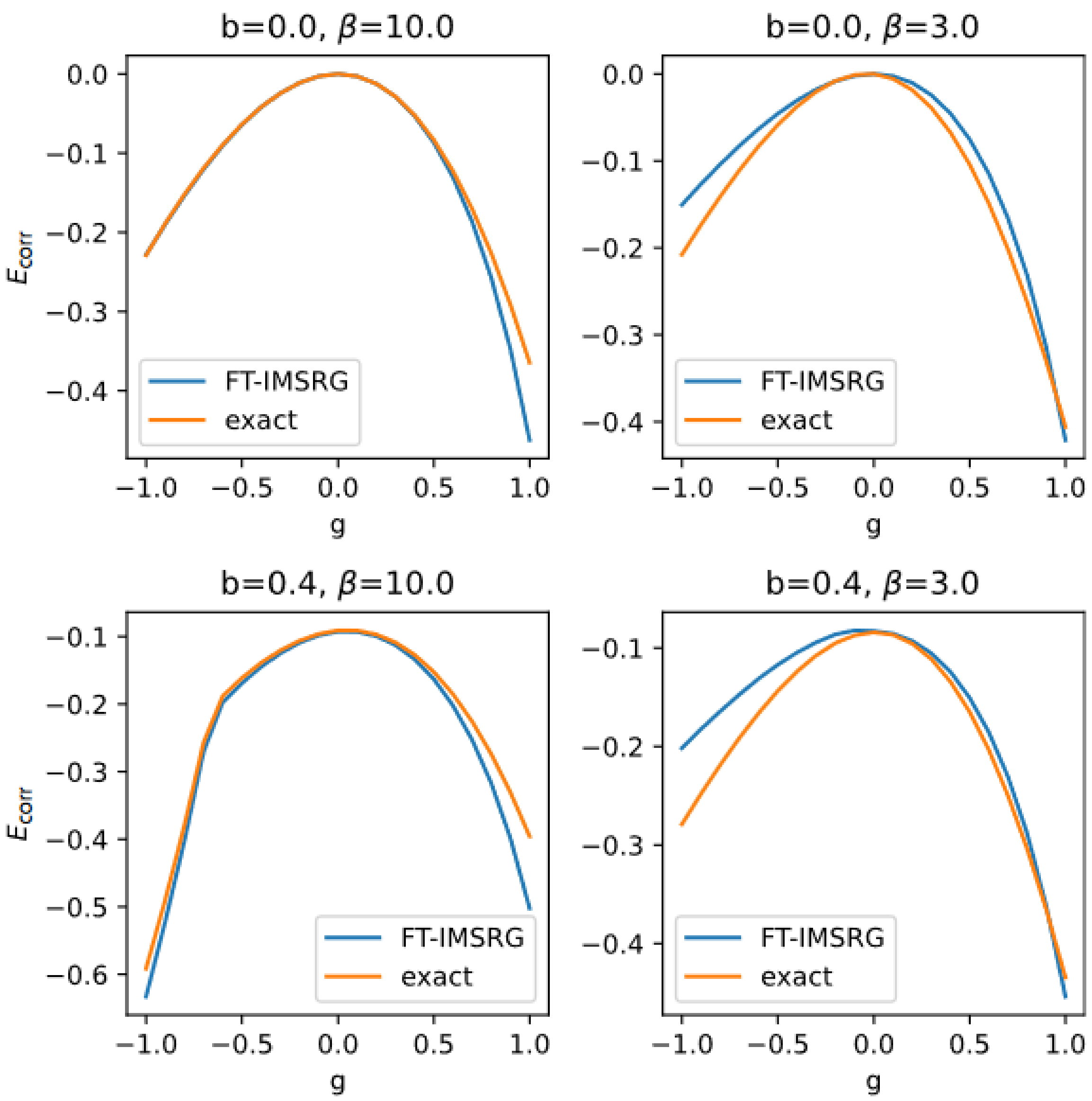


FIG. 4. Relative error in internal energy versus inverse temperature for $b = 0$ (top) and $b = 0.2$ (bottom) with various values of g on a logarithmic scale. The FT-HF results are shown with dotted lines and the FT-IMSRG results are shown with solid lines. In all cases, the FT-IMSRG results show a significant improvement compared to the FT-HF results, with better results for weaker coupling.

- 相比于FT-HF，在高温和低温情况下，FT-IMSRG更接近严格解。
- 底部图中不太平滑的结果可能是由于哈密顿量中的配对破坏项引起的数值复杂性。

Result



$$E_{\mathrm{corr}} = E - E_{\mathrm{HF}}$$

FIG. 5. Correlation energy versus g for various values of b and β. At low temperature, the FT-IMSRG results (blue) and exact results (orange) are in great agreement. Weaker coupling in both b and g produce more accurate FT-IMSRG results.

- 有趣的是，在较高温度下(右,β=3)，负g(吸引配对)的偏差更大；而在较低温度下(左,β=10)，正g(排斥配对)的偏差更为显著。
- 在低温时，系统更接近有序的基态，量子关联效应显著。对于排斥性配对（g>0），相互作用本身不利于形成关联的基态。
- 高温带来强烈的热涨落会削弱任何强烈的关联倾向。对于吸引性配对（g<0），高温会破坏这种有序的配对关联，使系统行为趋于无序。

Result

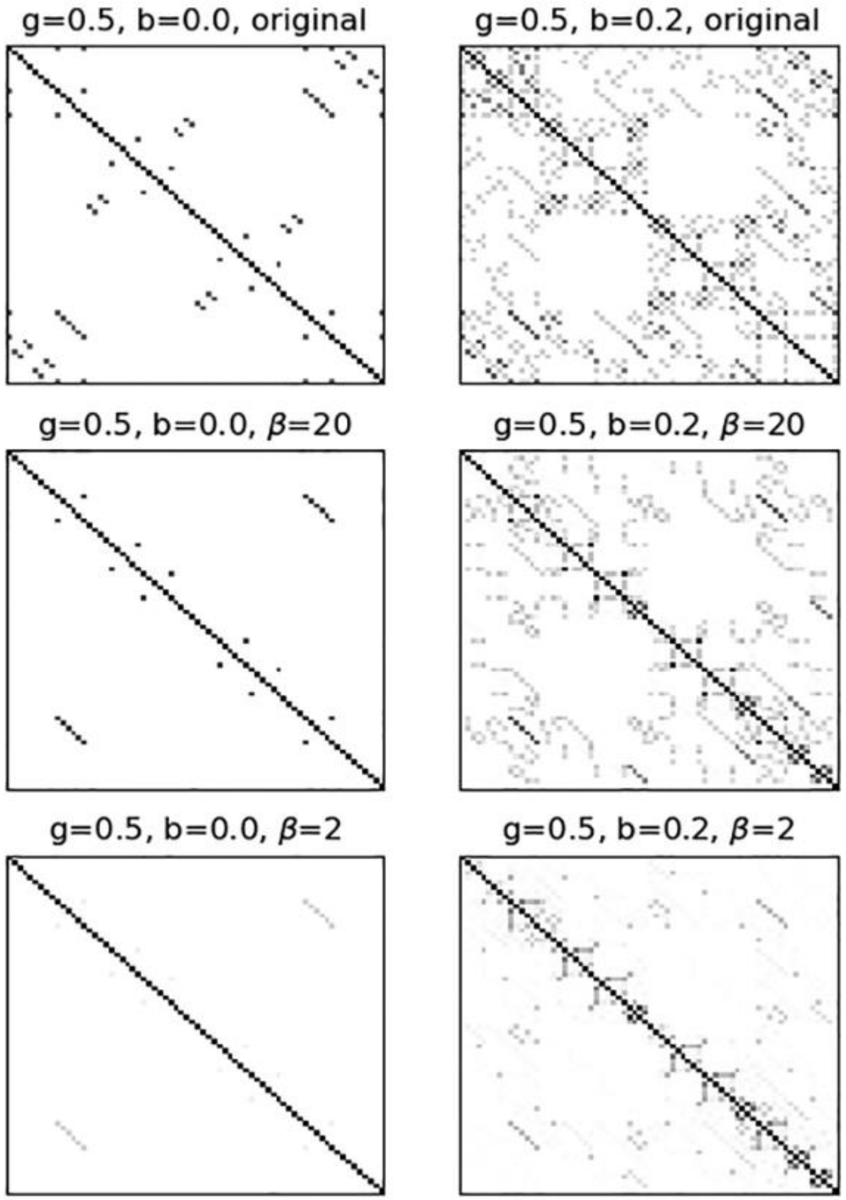


FIG. 6. The full 70×70 many-body Hamiltonian matrix for $A = 4$ and $N = 8$ and two sets of parameters. We compare the unevolved Hamiltonian at zero temperature ($\beta \to \infty$, row), to the FT-IMSRG evolved Hamiltonian for $\beta = 20$ (middle row), and for $\beta = 2$ (bottom row). Darker colors correspond to matrix elements with larger absolute values, and matrix elements with a value of zero are shown in white. At low temperatures, the FT-IMSRG decouples only the lowest-energy states, while at higher temperatures the FT-IMSRG decouples many more states.

- 在低温下，FT-IMSRG仅解耦最低能态(与零温IMSRG情况相同)，而在较高温度下，FT-IMSRG会解耦更多能态。但这是以截断误差为代价。
- 当β=20(极低温)时，纯配对的截断误差低于0.25%，加入破坏配对相互作用，截断误差则低于0.55%。
- 当β=3(高温)时，纯配对的截断误差低于0.86%，加入破坏配对相互作用，截断误差则低于3%。

Result

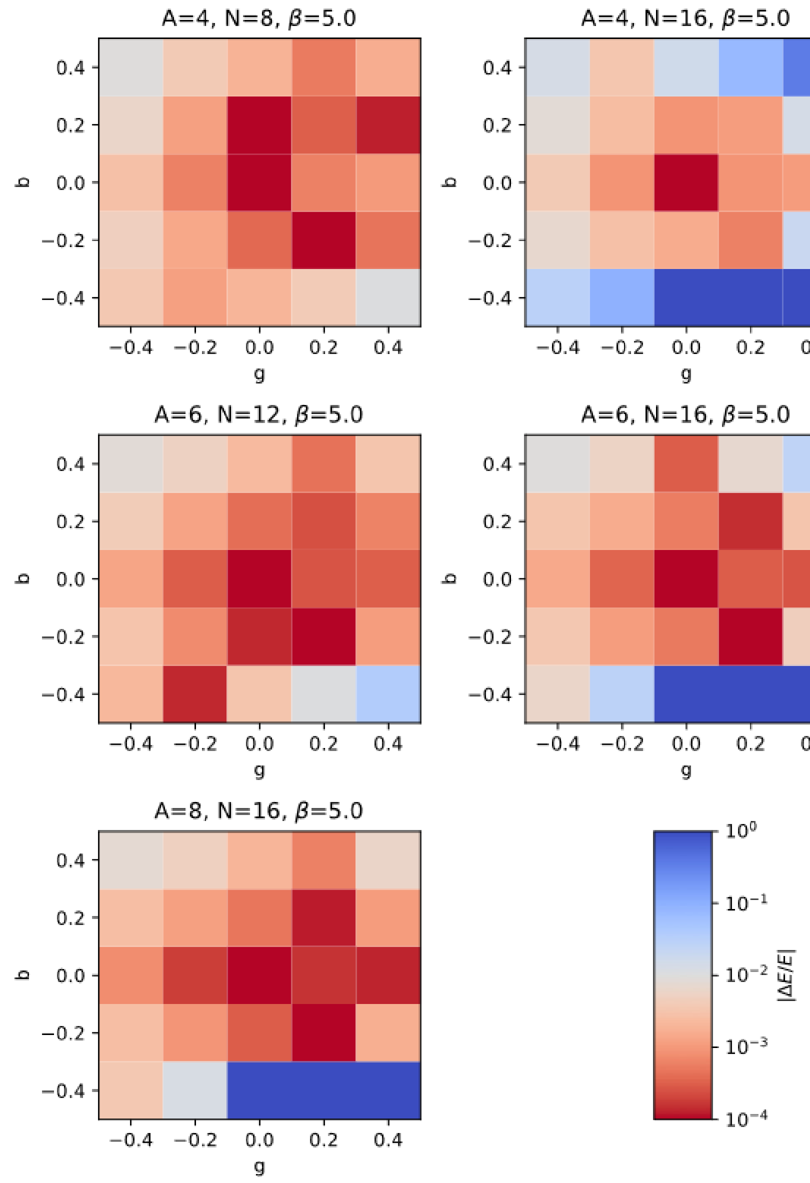


FIG. 7. Relative error in internal energy in parameter space for $\beta = 5.0$ and various values of A and N with a logarithmic color scale. Shades of red denote less than 1% error. The FT-IMSRG results display the most accuracy for higher A and lower N.

- 当固定A而增加N时，大多数参数的相对误差会略有上升。
- 需要注意的是，在将FT-IMSRG应用于原子核时，会通过固定A值增加N，以使结果在基矢尺寸方面达到收敛。

Result

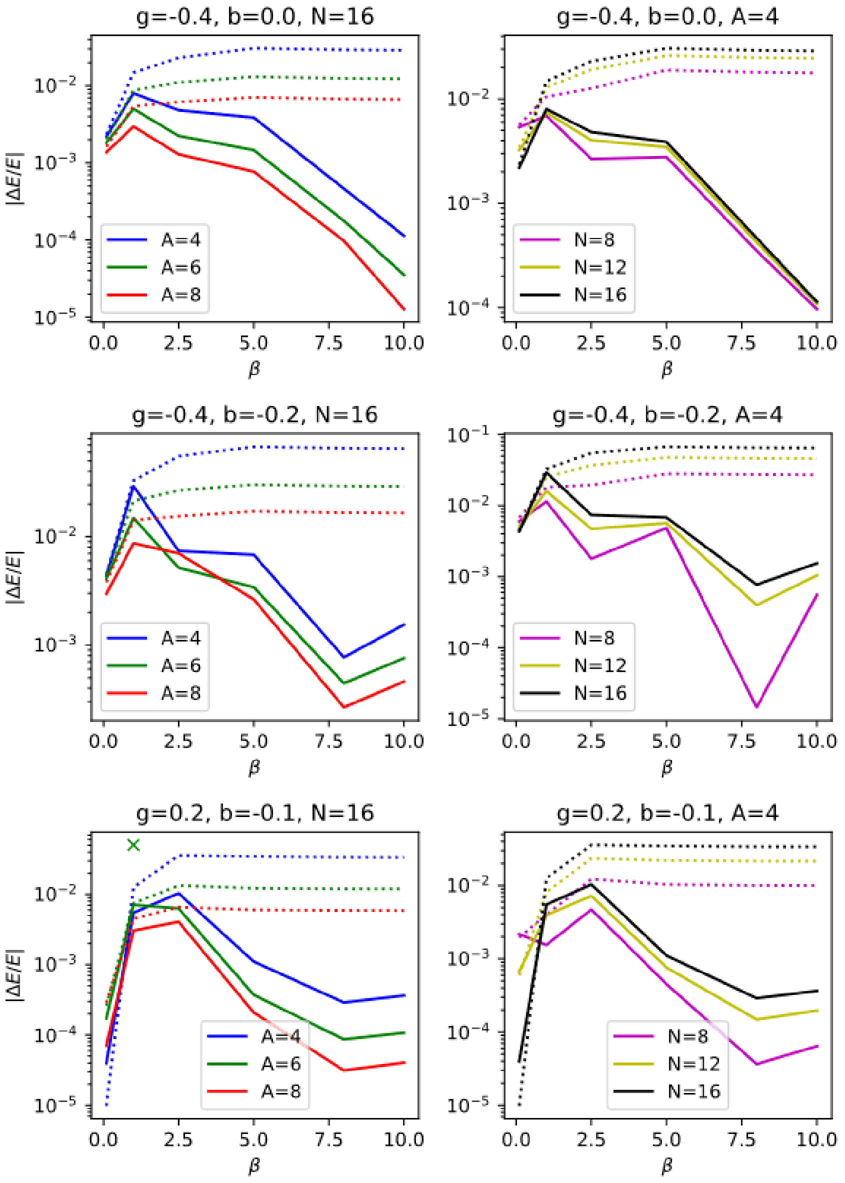


FIG. 8. Relative error in internal energy versus inverse temperature on a logarithmic scale for various values of g, b, A, and N. The FT-HF results are shown with dotted lines and the FT-IMSRG results are shown with solid lines. In the bottom left panel ($g = 0.2$, $b = -0.1$, $N = 16$), we encountered numerical instability at $A = 6$, $\beta = 1$. The results for neighboring values of β are plotted, with the unstable result at $\beta = 1$ marked with an x.

- 随着A的增加，FT-IMSRG的精度显著提高；而随着N的增加，其精度略有下降。
- FT-IMSRG方法通常能改进FT-HF的结果，但在高温(低β)的少数情况下例外，这可能是由截断误差引起的。
- 值得注意的是，右列中随着N的增加，误差似乎从下方收敛到一个固定值。

Result

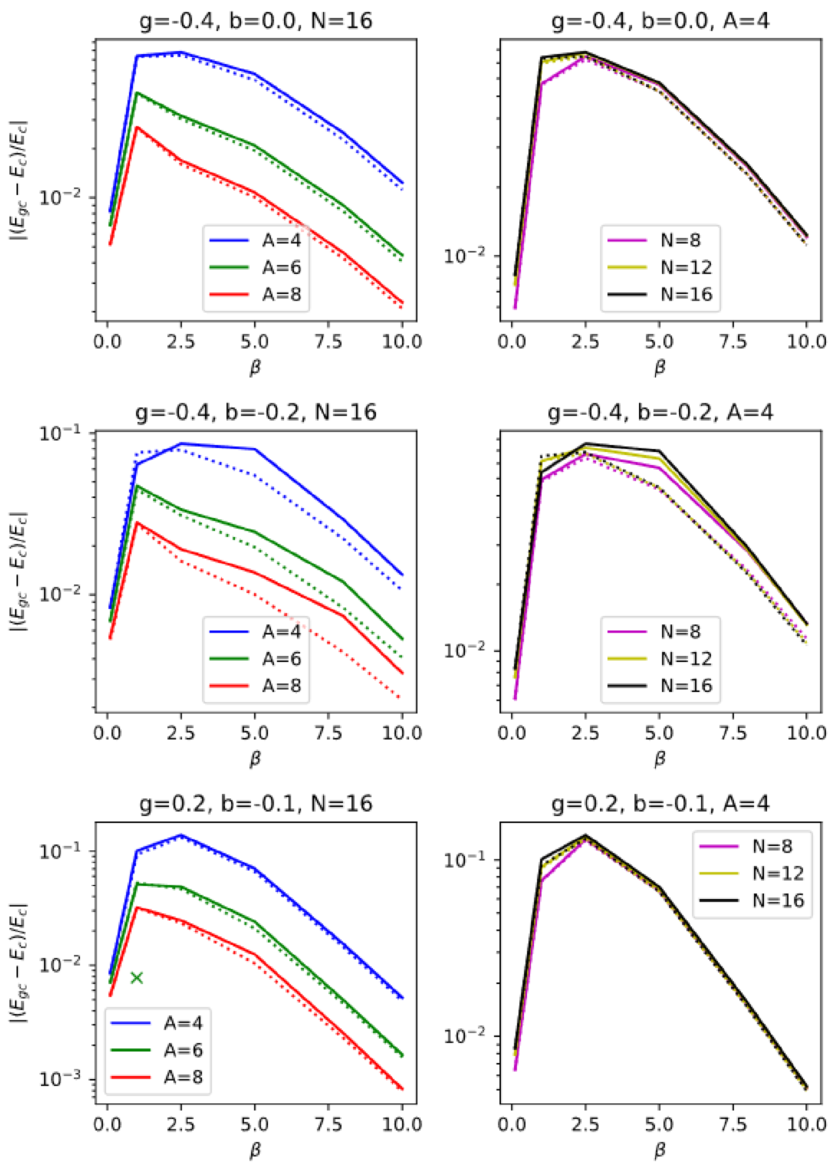


FIG. 9. Relative difference between the canonical ensemble and grand-canonical ensemble results versus inverse temperature on a logarithmic scale, using the same parameters as Fig. 8. The FT-HF results are shown with dotted lines and the FT-IMSRG results are shown with solid lines. As with Fig. 8, in the bottom left panel ($g = 0.2, b = -0.1, N = 16$) we encountered numerical instability at $A = 6, \beta = 1$ for the canonical ensemble. The results for neighboring values of β are plotted, with the unstable result at $\beta = 1$ marked with an x.

- 随着A的增加，各系综之间的差异明显减小。这是因为在多粒子情况下，粒子涨落带来的影响更小。
- 改变N对整体之间的差异影响要小得多。
- 对于粒子数A ≥ 8的系统，使用巨正则系综似乎是安全的；但对于粒子数较少的系统，这种近似确实会引入1%-10%范围内的误差。

Result

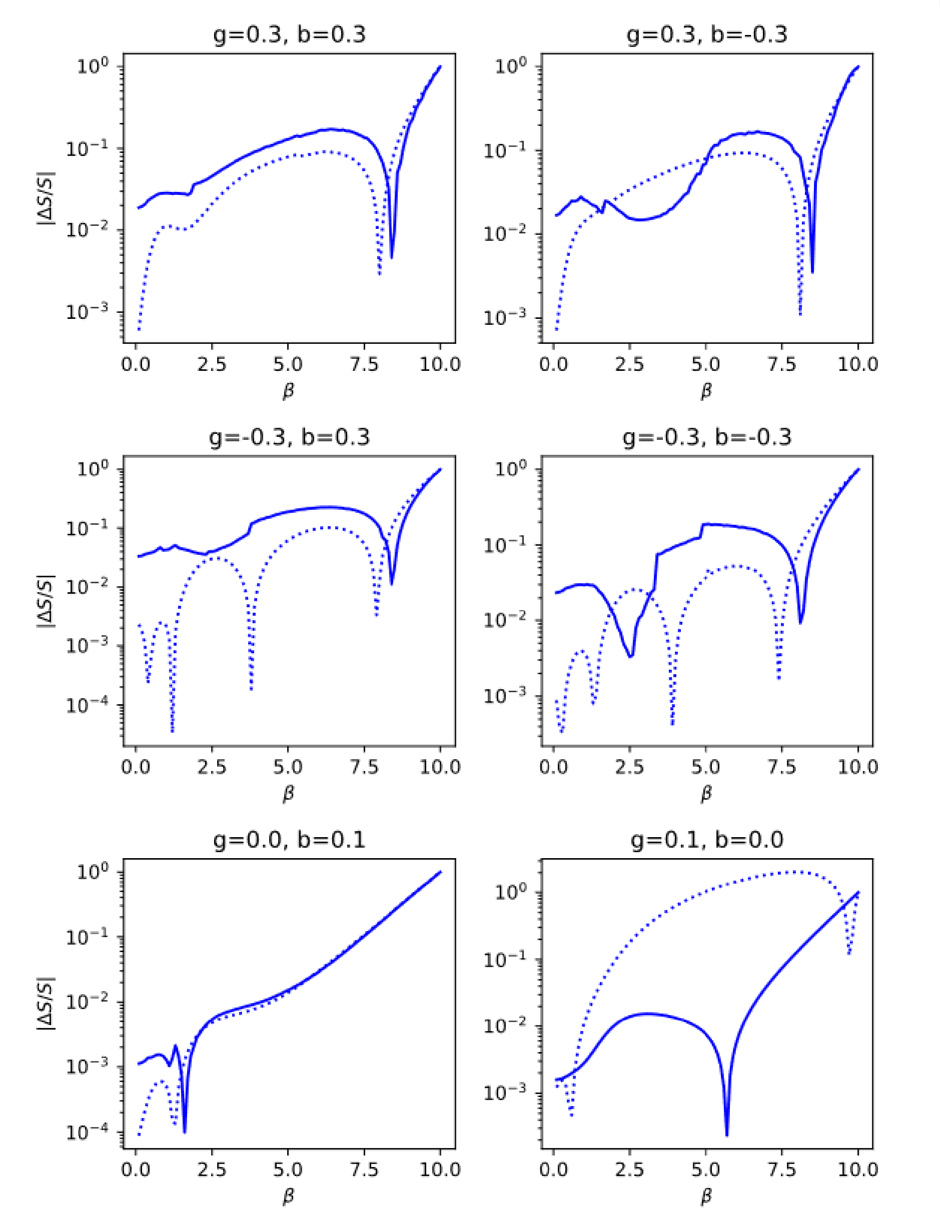


FIG. 10. Relative error in entropy versus inverse temperature for $A = 4$ and $N = 8$ with various values of g and b on a logarithmic scale. The FT-HF results are shown with dotted lines and the FT-IMSRG results are shown with solid lines. The 100% error at $\beta = 10$ is a result of assuming $E(\infty) = E(10)$ in the integration limits of Eq. (80). Entropy results are more accurate for weaker coupling, and the difference between FT-IMSRG and FT-HF in terms of entropy is generally small.

$$S(\beta) = \int_{E(\infty)}^{E(\beta)} \beta'(E')dE'.$$

- 熵的结果在耦合较弱时更为准确，而FT-IMSRG与FT-HF在熵方面的差异通常较小。

Result

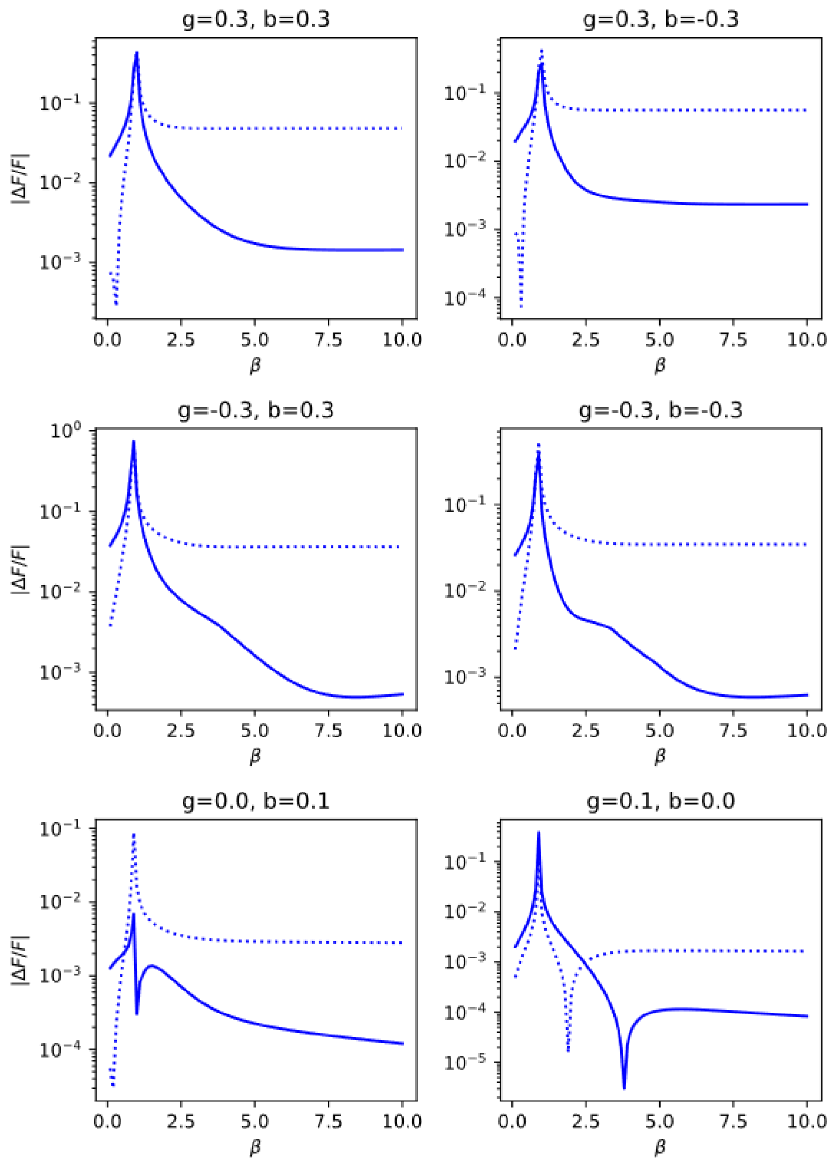


FIG. 11. Relative error in Helmholtz free energy versus inverse temperature for $A = 4$ and $N = 8$ with various values of g and b on a logarithmic scale. The FT-HF results are shown with dotted lines and the FT-IMSRG results are shown with solid lines. Beyond $\beta \approx 2$, the FT-IMSRG results are consistently more accurate than the FT-HF results.

$$F = E - TS.$$

- FT-IMSRG 对自由能的计算明显优于 FT-HF 的结果，因为 F 的准确性很大程度上取决于内能计算的精度。因此，F 的相对误差与内能的相对误差相似是合理的。

Conclusion

- 我们发现FT-IMSRG与精确解在低温下和哈密顿量弱耦合时吻合最佳，但FT-IMSRG在广泛的参数和温度范围内都能产生高度精确的结果，优于FT-HF方法。
- 在我们研究具有不同粒子数A和单粒子态数量N的P3H模型时，发现FT-IMSRG的计算精度随着A的增加而提高。在构建FT-HF占据数时，选择正则系综还是巨正则系综会对FT-IMSRG结果产生显著影响，但这种影响会随着粒子数的增加而明显减弱。
- 作为下一步工作，我们将采用基于手征有效场理论推导的核相互作用力，对原子核进行FT-IMSRG计算。重点研究中子滴线等核结构特征随温度升高的演化规律，并计算与恒星环境核过程（包括核合成）相关的反应速率和衰变速率。
- 与此同时，我们将继续在其他IMSRG变体及基于IMSRG的结合方法中实现有限温度的应用。

Thanks!